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John H. Hall, Simeon North, and the Milling Machine: The Nature of Innovation among Antebellum Arms Makers

MERRITT ROE SMITH

During the early decades of the 19th century, the firearms industry in the United States experienced a remarkable transformation from craft to machine production. The origins of this change can undoubtedly be traced to Europe, particularly France and Great Britain. Nevertheless, Americans made some boldly original innovations of their own and, in so doing, sowed the seeds for a system of interchangeable production that eventually bore their name. Indeed, the continuous introduction of novel wood- and metal-working techniques at various armories throughout the country not only laid the basis for precision manufacture in other technically related industries, but also, in the eyes of European observers, represented one of the most exhilarating features of Yankee technology during the antebellum period. Exciting envy and admiration, they became great jewels in America's industrial crown.¹

The emergence of modern milling practice provides an apt illustration of the migration and diffusion of mechanical ideas during the formative period of American industrialization.² Though by no means the only requisite for interchangeable manufacture, the mill-

DR. SMITH, assistant professor of history at Ohio State University, conducted the research for this study as a predoctoral fellow of the Smithsonian Institution. An earlier version of this paper was read at the annual meetings of the Society for the History of Technology and the American Historical Association, New York City, December 30, 1971. The author wishes to thank those who offered criticism at that time, particularly Edwin A. Battison, who generously shared unpublished research information and subsequently rendered valuable technical advice.

¹Over the years a number of scholars have provided able analyses of the so-called American System. The most recent is Nathan Rosenberg, ed., *The American System of Manufactures* (Edinburgh, 1969).

²Though it is too early to make any definite pronouncements, it appears that the concept of milling iron had a European ancestry which predated American developments.

ing machine contributed significantly to the system's rapid maturation and ultimate success. In fact, so far as the first glimmerings of interchangeable production accompanied the successful introduction of milling machinery, it is not far from the mark to say that the story of modern milling not only sheds light upon the transfer of technology during the early 19th century but also presents an encapsulated overview of the rise of the American System.

By mechanically feeding a table holding the workpiece (or part to be cut, shaped, or smoothed) into a rotary multiple-toothed cutter, the milling machine achieved a degree of precision far surpassing anything that could be accomplished by hand filing the same component in a jig.³ Although considerations of time and increased efficiency ultimately favored its adoption by cost-conscious arms-making establishments, an ever increasing demand for more uniform small arms by the U.S. Ordnance Department initially prompted experimentation with milling metallic substances and, for that matter, a whole host of labor-saving machine tools and gauging devices that began to appear after the War of 1812.⁴ Given this context, three questions are of paramount concern: What inventors contributed to the development of the modern concept of milling? Can meaningful links be established between these men? And, finally, what does this information reveal about the nature of innovation among early American mechanics?⁵

* * *

If anyone deserves credit for conceiving and developing the milling machine in America, it is Simeon North, a soft spoken but singularly talented small-arms contractor from Middletown, Connecticut.⁶

³A clear-cut distinction between true milling and other machining techniques is provided in Edwin A. Battison's account of "Eli Whitney and the Milling Machine," *Smithsonian Journal of History* 1 (1966): 9, 16–23.

⁴See, for example, Roswell Lee to James Stubblefield, August 6, 1816, Record Group 156, Records of the Springfield Armory, Letters Sent, National Archives, Washington, D.C. (hereafter cited as RG 156, SAR); Lee to Senior Officer of Ordnance, November 20, 1817, Record Group 156, Records of the Office of the Chief of Ordnance, Letters Received, National Archives (hereafter cited as RG 156, OCO); Colonel Decius Wadsworth to John C. Calhoun, October 13, 1818, RG 156, OCO, Letters Sent to the Secretary of War; Roswell Lee to Asa Waters, June 8, 1819, RG 156, SAR, Letters Sent; Colonel George Bomford to Lee, September 21, 1821, RG 156, OCO, Letters Sent; Bomford to Lee, September 3, 1821, *ibid*.

⁵The answers I would like to suggest are derived primarily from undisclosed manuscripts housed at the National Archives.

⁶Simeon North (1765–1852) had a very long and prolific career. Born and raised on a farm near Berlin, Connecticut, he began manufacturing scythes and other agricul-

Several pieces of evidence establish the date well before 1818. In 1900, Edward G. Parkhurst, a former resident of Middletown and long time engineer with Pratt and Whitney of nearby Hartford, described and illustrated an old milling machine he had seen in 1851 at Robert Johnson's neighboring "Mill Hollow" arms works in Middletown. Though Johnson "did not claim it was his own invention," he "said it was the first he had ever known of" and "gave the year in which it was first put to work as 1818."⁷ As the machine performed only small flat cuts, operated by means of a hand-cranked rack and pinion feed, and "had no adjustment whatever," it lacked several features possessed by later machines at Harpers Ferry, Virginia, Whitneyville, Connecticut, and Springfield, Massachusetts. Yet, though primitive by modern standards, it nonetheless performed a true milling function.

It is also noteworthy that in 1829 another private arms maker from Middletown, Nathan Starr, had a "Milling Engine" similar to the one described by Parkhurst except that it possessed an extra spindle support and employed a mill cutter "made of properly tempered steel having its surfaces obliquely furrowed, and of a form corresponding to the shape to which the iron is to be reduced."⁸ With the use of this machine as well as a "Chipping Engine" for shaping iron components, Starr declared to have "made a complete and perfect revolution" in the manufacture of arms insofar as he dispensed with the use of jigs and saved "an immense expense in files." Furthermore, he claimed to have invented these machines and used them at his factory "some years" before 1829.⁹ However, as Starr was

tural implements in 1795. Several years later he turned his attention to making firearms and in 1799 secured his first government contract for 500 horsemen's pistols. By 1813 the business had grown to such an extent that North moved his family to Middletown and erected a new \$100,000 factory on Staddle Hill. While continuing the manufacture of pistols, he undertook the production of military rifles for the United States in 1823 and five years later signed an \$87,500 contract for 5,000 Hall breech-loading rifles. From that time until his death in 1852 North devoted his efforts almost exclusively to the manufacture of Hall rifles and carbines. For further information see S. N. D. North and Ralph H. North, *Simeon North; First Official Pistol Maker of the United States* (Concord, N.H., 1913); and the *Dictionary of American Biography*, s.v. "North, Simeon."

⁷Edward G. Parkhurst, "One of the Earliest Milling Machines," *American Machinist*, March 8, 1900, p. 217. See also George A. Stockwell, *The History of Worcester County, Massachusetts* (Boston, 1879), p. 113.

⁸In an earlier draft of this paper I neglected to indicate that Starr's machine differed from North's in that it had three spindle supports instead of two. I am grateful to Edwin A. Battison for calling attention to this discrepancy.

⁹Nathan Starr to John H. Eaton, July 10, 1829, RG 156, OCO, Special File, Correspondence Relating to Inventions. In addition to his engines for milling and chipping

attempting to renew his contract with the government in 1829, it appears that he made this claim primarily to impress ordnance authorities of his ability to fulfill an obligation.¹⁰ Whatever his intention, he did not patent these machines or prove conclusively that they originated at his works.

The situation is quite otherwise with Simeon North. While attempting to renew his government contract for Hall carbines in 1852, the old armorer contended that "he first invented the mode of milling iron . . . about the year 1816."¹¹ That this date is substantially correct seems verified by the fact that Roswell Lee and James Stubblefield, superintendents at the Springfield and Harpers Ferry national armories, respectively, made a special trip to Middletown in March 1816 to examine several mechanical improvements introduced by Colonel North in manufacturing pistols for the public service.¹² Moreover, in 1852 North produced sworn affidavits affirming his inventive priority from his neighbor and fellow arms maker, Robert Johnson; his shop foreman, Selah Goodrich; and a former workman named Joseph Weatherhead. Although Weatherhead emphasized mainly the colonel's work with interchangeable parts, both Johnson and Goodrich specifically mentioned the milling machine.¹³ According to Johnson, North was "the first person, who to my knowledge, milled his work generally in carrying on the business of a gunsmith."¹⁴ Goodrich not only agreed, but emphatically "repudiated the suggestion that they copied anything from [Eli] Whitney."¹⁵ The evidence

iron, Starr included descriptions of two machines for turning and rifling gun barrels. Both machines, however, were well known and used at other armories prior to 1829.

¹⁰Colonel George Bomford to Nathan Starr, June 22, 1829, RG 156, OCO, Letters Sent; Bomford to Starr, January 19, 1830, *ibid*.

¹¹Statement of Simeon North, Middletown, Connecticut, April 20, 1852, RG 156, OCO, Letters Received; Colonel Henry K. Craig to Charles M. Conrad, April 23, 1852, *ibid*.

¹²Roswell Lee and James Stubblefield to Senior Officer of Ordnance, March 20, 1816, RG 156, SAR, Letters Sent. In addition to inspecting special machinery devised by North for making gunstocks and turning barrels, Lee and Stubblefield specifically mentioned that he had "made an improvement in the locks by fitting every part to the same lock, which ensures a more rigid uniformity than they have here to-fore known." Given the purpose of their mission, this may have been a reference to North's milling machine.

¹³Joseph Weatherhead to Simeon North, April 12, 1852, RG 156, OCO, Letters Received. Upon leaving North's works in 1818, Weatherhead went to the Springfield armory where he worked as a machinist and served as master armorer on three different occasions: 1825–33, 1843, and 1844–47.

¹⁴Statement of Robert Johnson, Middletown, Connecticut, April 5, 1852, RG 156, OCO, Letters Received.

¹⁵Statement of Selah Goodrich, Middletown, Connecticut, April 1, 1852, RG 156, OCO, Letters Received; Charles H. Fitch, "The Rise of a Mechanical Ideal," *Magazine of American History* 11 (June 1884): 520.

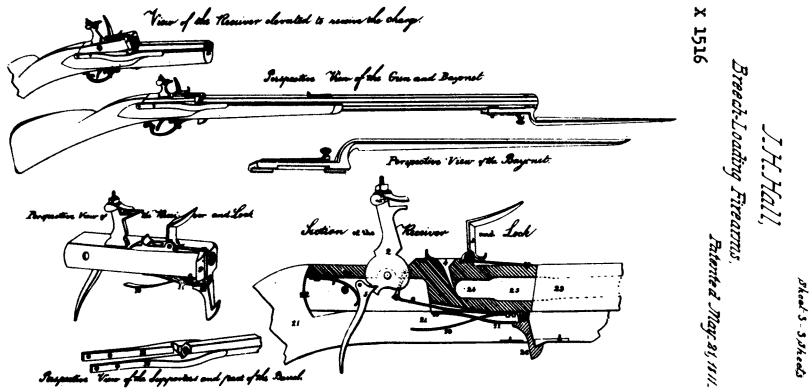


FIG. 1.—Original drawing of Hall's breech-loading rifle submitted to the United States Patent Office in the spring of 1811. Though Hall did not design this rifle for interchangeable production, by 1819 he had abandoned the original pattern and developed a model that could be prepared by machine processes. (United States Patent no. 1516, May 21, 1811, Department of Commerce.)

strongly indicates, then, that the earliest known milling machine in America originated at North's factory and that the owner devised it.

* * *

A second stage in the development of modern milling methods occurred at the Harpers Ferry armory in Virginia between 1819 and 1826. A transplanted New Englander named John H. Hall was the innovator. A native of Portland, Maine, Hall arrived at Harpers Ferry in the spring of 1818 to construct experimental models of a breech-loading rifle he had copatented seven years earlier with the superintendent of patents, William Thornton (see fig. 1). With crucial support from the Maine delegation to Congress, he succeeded in getting the endorsement of Secretary of War John C. Calhoun and in March 1819 received a contract to manufacture 1,000 rifles upon the "uniformity principle," that is, on what we now refer to as interchangeable parts.

As it took over five years to tool his shop and complete the contract, a number of officials at Harpers Ferry and Washington expressed skepticism about Hall's ability to produce rifles inexpensively with interchangeable parts.¹⁶ Consequently, in response to a congressional

¹⁶Edward Lucas, Jr., to John Cocke, April 27, 1826, RG 156, OCO, Letters Received; Thomas Copeland to Major William Wade, May 29, 1826, *ibid.*; Colonel George Bomford to James Barbour, July 3, 1826, RG 156, OCO, Letters Sent to the Secretary of War; Bomford to John H. Hall, November 16, 1826, RG 156, OCO, Letters Sent; Hall to Bomford, December 9, 1826, RG 156, OCO, Letters Received; Edward Lucas, Jr., to John Cocke, January 15, 1827, *ibid.*

resolution calling for information on the subject, the Ordnance Department appointed a three-man committee headed by James Carrington of Wallingford, Connecticut, to examine Hall's machinery in December 1826.¹⁷ After spending three weeks at the rifle works, the members of the commission assembled their notes and prepared a seventeen-page report which they submitted to the secretary of war early in January 1827.¹⁸

It is doubtful whether Hall himself could have composed a more laudatory document. Members of the Artillery School of Practice at Fortress Monroe had praised his rifle (see figs. 2 and 3), but the Carrington committee found his machinery unparalleled in contemporary practice. "Capt. Hall," they asserted, "has formed & adopted a system, in the manufacture of small arms, entirely *novel* & which no doubt, may be attended with the most beneficial results to the Country, especially, if carried into effect on a large scale." Continuing, they pointed out that "his machines, for this purpose, are of several distinct classes, and are used for cutting iron & steel & for executing wood work; all of which are essentially different from each other & differ materially from any other machines, we have seen, in any other establishment."¹⁹

From the tenor of the committee's remarks one might infer that Hall conceived his machinery independent of any outside influences. Such, however, is not the case. Hall was evidently well aware of Simeon North's accomplishments, since North had visited Harpers Ferry in 1816 for the express purpose of introducing his "uniform plan of work" at the national armory.²⁰ Given the date of its invention, it is quite likely North conveyed knowledge of his milling machine

¹⁷Carrington possessed impressive credentials. Well known and respected as an armorer as well as a machinist, he had served as foreman of Eli Whitney's celebrated "Gun Manufactory" near New Haven between 1799 and 1825. The second member of the commission, Luther Sage, was an inspector of contract arms at Springfield and Philadelphia and thoroughly familiar with the works of Simeon North, Nathan Starr, and Robert Johnson of Middletown. The third member, Colonel James Bell of West Brook, Virginia, was a last-minute replacement for Gideon Davis, a millwright and foundry owner from Georgetown, D.C. He admittedly knew very little about mechanical subjects. Questions concerning the criteria used to select this committee and the objectivity of its findings are most interesting and the subject of historical debate. Unfortunately, however, they go beyond the immediate scope of this paper.

¹⁸James Carrington, Luther Sage, and James Bell to Colonel George Bomford, January 6, 1827, RG 156, OCO, Special File, Reports and Correspondence of Ordnance Boards. This is the original report and is written in Carrington's hand (hereafter cited as the Carrington Committee Report).

¹⁹*Ibid.*

²⁰James Stubblefield to Roswell Lee, May 9, 1816, RG 156, SAR, Letters Received. The author is indebted to Edwin A. Battison for providing this reference.

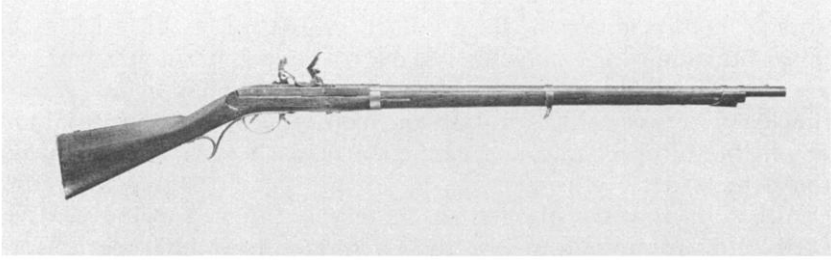


FIG. 2.—Model 1819 Hall rifle manufactured at Harpers Ferry in 1824. In July 1826 two companies of the Artillery School of Practice at Fortress Monroe, Virginia, were armed with weapons of this vintage. After five months of extensive tests the officers in charge of the operation submitted a lengthy report which not only testified to their superiority “over every other Small Arm now in use” but acknowledged the interchangeability of their component parts. (Courtesy of the Division of Military History, Smithsonian Institution.)

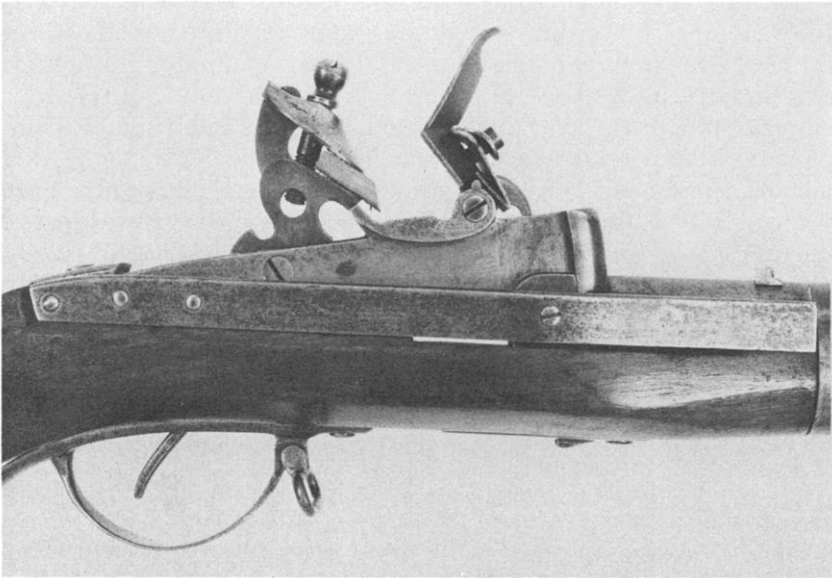


FIG. 3.—Detail of the breech mechanism of Hall's rifle dated 1824. (Courtesy of the Division of Military History, Smithsonian Institution.)

to Harpers Ferry, perhaps even in the form of sketches or drawings. Further, it is known that Hall spent some time at the Springfield armory before going to the Potomac in 1818. He may well have passed through Middletown, seen North's machinery in operation, and derived direct inspiration from it prior to devising his own at the Ferry between 1821 and 1825. Such visits were quite common among master mechanics at the time, particularly when they practiced the same trade.

Although Hall constructed machinery for drop forging components and profiling gunstocks, the Carrington committee paid special attention to his metal-cutting engines. Installed at the rifle works during the early 1820s, these machines performed with cutters and saws "work usually done elsewhere with grindstones, chisels & files."²¹ The principles on which these contrivances operated, however, did not originate with John Hall. For instance, the Carrington report mentions a trimming machine used for cutting off the surplus metal of forgings. Though constructed "to operate with extreme *Steadiness* & almost *unlimited* power," it definitely resembled a machine used at Springfield for similar purposes since 1818.²² The same held true for a number of vertical spindle machines used at the rifle works for hollow milling screw pins as well as drilling, reaming, and counter-sinking holes in components.²³

In addition to standard units for hollow milling and trimming components, Hall patented several other machines in 1827 "for cutting metallic substances."²⁴ Because drawings of these contrivances have yet to be found, their exact specifications remain hazy. Nevertheless, the Carrington committee designated them as "*Straight cutting, Curved cutting & Lever cutting Machines*."²⁵ Of the three, Hall's straight cutting

²¹Carrington Committee Report, January 6, 1827, OCO; Charles H. Fitch to James H. Burton, July 23, 1881, Burton Papers, Yale University Library, New Haven, Conn.: Charles H. Fitch. *Tenth Census of the United States: Manufactures*, 22 vols. (Washington, D.C., 1883), 2:636.

²²Roswell Lee to Colonel Decius Wadsworth, January 10, 1818, RG 156, OCO, Letters Received.

²³Colonel John Whiting to William Eustis, January 13, 1810, Record Group 107, Records of the Office of the Secretary of War, Letters Received (Main Series), National Archives, Washington, D.C.; *American State Papers: Military Affairs*, 7 vols. (Washington, D.C., 1832), 1:551; Battison (n. 3 above), pp. 16–26.

²⁴Edmund Burke, *List of Patents for Inventions and Designs Issued by the United States from 1790 to 1847* (Washington, D.C., 1847), p. 51; M. D. Leggett, *Subject-Matter Index of Patents . . . from 1790 to 1873*, 3 vols. (Washington, D.C., 1874), 2:922.

²⁵Carrington Committee Report, January 6, 1827, OCO.

Hall's "curved cutting" machine served "to produce surfaces of either *double* or *single curvature* of any of the regular curves." At the time the commissioners made their examination it had been fitted to produce only circular forms. Years later James H.

machines are most interesting and historically important because they represent an early form of plain milling as we know it today. Designed “to produce surfaces either *straight* and *flat* or *straight & fluted* or *ribbed*,” the commissioners thought they could easily be “applied to the production of a great variety of other surfaces both regular & irregular.”²⁶ From this it is clear that they not only employed formed cutters but also must have incorporated several improvements not possessed by North’s earlier milling machine.²⁷

The most distinctive feature of Hall’s cutting engines is that he employed “common hands” and “even boys” to tend them without the slightest fear of jeopardizing standards of accuracy or the quality

Burton tentatively identified it as a “bridge-milling machine or a lathe with a former.” The “lever” machine executed such work as could not “be so conveniently effected by either of the other machines.” Applied principally to the breech mechanism of Hall’s rifle, in December 1826 its functions included boring the pan and cutting recesses in the receiver for the battery, cock, and main spring. Writing to James H. Burton in 1881, Charles H. Fitch described it as “similar in principle to some forms of hand-milling machines now in use.” See Fitch to Burton, July 23, 1881, Burton Papers (n. 21 above).

²⁶Ibid.

²⁷Powered by water, Hall’s cutting machines operated by means of leather belts which, “by increasing their *tension & width*,” overcame “any resistance, however great, arising from their number or the velocity of their motions.” Specifically, the Carrington Committee observed that the numerous pulleys revolved “without any of that *shaking & trembling*” which characterized machines used elsewhere. This was accomplished by balancing them with lead after they had been finished in the lathe. The importance of having balanced pulleys cannot be overestimated because it enabled Hall to run his machines at much faster speeds than usual without sacrificing steady accurate performance. In 1829, Inspector General John E. Wool remarked that they not only operated efficiently at 3,000 revolutions per minute but also were constructed so the speed of the moving parts could “be either increased or diminished at discretion” (Wool to General A. Macomb, October 31, 1829, Record Group 159, Records of the Office of the Inspector General, Inspection Reports, National Archives).

Another factor enhancing the effectiveness of Hall’s cutting machines was their rugged size and durability. Nearly every person who inspected the rifle works during the 1820s and 1830s perceived the relationship between their stability and accurate performance. On more than one occasion ordnance officers spoke highly of their “massive style of construction” (Colonel George Talcott to Colonel George Bomford, December 15, 1832, RG 156, OCO, Special File, Reports of Inspections; Talcott, Inspection of the Hall Rifle Works, October 31, 1836, *ibid.*). And though he considered them “very clumsy” by later standards, even Charles H. Fitch recognized the significance of Hall’s “excessively solid and heavy” machines (Fitch, “The Rise of a Mechanical Ideal” [n. 15 above], p. 519).

Perhaps the best testimony to the durability and utility of Hall’s machinery comes from the armory records themselves. With the delivery of the last parcel of patent rifles in 1844, the rifle works housed forty-nine cutting machines excluding those used for trimming, drilling, and hollow milling. Under orders from the superintendent, master armorer Benjamin Moore, James H. Burton, and several other workmen

of work performed.²⁸ "To keep a machine in operation," the Carrington committee observed, "*activity* is more necessary than *judgement*, for the machines, after the work is put into them, go thro' with the operation without any further *aid* from the boy, and when the operation is completed, give notice to the boy, who has been employed during the operation, in putting in and taking out work from other machines."²⁹ In other words, Hall's cutting engines not only functioned without any manual guidance but evidently ceased operation once the workpiece had been finished. This enabled a laborer to tend as many as three or four machines simultaneously; and, as their management required "*no* mechanical skill," any reasonably alert individual could learn the job within a relatively short period of time.³⁰ Confident of their superiority over conventional methods, Hall candidly claimed that "one boy by the aid of these machines can perform more work than ten men with files, in the same time, and with greater accuracy."³¹ Though attempting to discredit their efficiency, one of Hall's sternest critics, Edward Lucas, acknowledged that one machine "completed as much work in a day, as three men could file by hand."³² The Carrington committee corroborated this

readjusted their counter pulleys and fitted them with new holders and cutters adapted to the production of components for the new model 1841 percussion rifle. With the exception of five units broken up for scrap in 1852, these machines continued in operation at Harpers Ferry up through 1855 and possibly as late as the Civil War. At face value, then, this evidence not only documents their remarkable longevity of service but amply corroborates Hall's claim that they possessed "universal application" and could be used equally well for the manufacture of common arms as for patent rifles (Major John Symington to Colonel George Talcott, September 30, 1845, RG 156, OCO, Letters Received; Symington to Talcott, December 9, 1847, *ibid.*; Colonel Benjamin Huger to Colonel Henry K. Craig, September 20, 1852, *ibid.*; John H. Hall to Joel R. Poinsett, February 21, 1840, Record Group 233, Records of the U.S. House of Representatives, Petitions, HR26A, hereafter cited as RG 233, HR).

²⁸John H. Hall to Colonel George Bomford, December 30, 1822, RG 156, OCO, Letters Received; Hall to Bomford, December 17, 1824, *ibid.*; Hall to John C. Calhoun, December 21, 1824, *ibid.*; Hall to Bomford, October 11, 1834, *ibid.*; Colonel George Talcott, Inspection of the Hall Rifle Works, October 31, 1836, OCO.

²⁹Carrington Committee Report, January 6, 1827, OCO.

³⁰John H. Hall to Daniel Bedinger, April 19, 1835, RG 156, OCO, Letters Received; Captain J. F. Lee to Colonel George Talcott, February 10, 1844, *ibid.*

³¹John H. Hall to Joel R. Poinsett, February 21, 1840, HR.

³²"But," Lucas hastily added, "this operation is not a fair test, because the receiver, upon which the trial was made is necessarily forged much larger, for the saws of the machine to act upon, than the same limb would be if forged for the purpose of being filed by hand." It is clear upon close examination of this statement that Lucas unknowingly contradicted himself; for, rather than exposing the supposed inefficiency of Hall's engines, he actually indicates that they were capable of removing a more substantial amount of iron in less time than could be accomplished by hand filing. See Lucas to John Cocke, January 15, 1827, OCO.

estimate, adding that “the cost of the cutters & saws does not amount to *half that* of files used to do the same quantity of work.”³³

According to James H. Burton, a former machinist and master armorer at Harpers Ferry, the Hall rifle works housed “not an occasional machine, but a plant of milling machinery by which the system and economy of the manufacture was materially altered.”³⁴ Featuring solid metal construction, formed cutting tools, variable speed control, and automatic stop mechanisms, Hall’s cutting machines represented a significant step forward in milling iron. Combined with sophisticated gauging techniques, they enabled him to dispense entirely with the use of jigs and develop a system of production capable of manufacturing arms with interchangeable parts.³⁵ In so doing, he far surpassed the legendary work of Eli Whitney and further elaborated upon that of Simeon North. Yet all this would have little historical significance if, as some scholars contend, Hall’s improvements had languished at Harpers Ferry.³⁶

* * *

Determining the extent of Hall’s influence is not a simple task. Essentially it means asking whether or not other arms makers adopted his techniques. On this question Hall’s sentiments are poignantly clear. Throughout the 1830s he asserted that his patented machinery had not only been introduced at the Springfield armory but also, by order of the Ordnance Department, used by Simeon North in manufacturing breech-loading rifles for the United States (see fig. 4).³⁷ His lawyer, Isaac N. Coffin, and immediate family made similar

³³Carrington Committee Report, January 6, 1827, OCO.

³⁴Charles H. Fitch to James H. Burton, July 23, 1881, Burton Papers (n. 21 above).

³⁵Though unable to apply the test to earlier specimens, I have examined a number of Hall rifles made between 1832 and 1838 and found that their major components do in fact interchange. Hall’s contemporaries clearly acknowledged this fact. They also pointed out, however, that the particular design of the rifle lent itself more readily to interchangeable manufacture than did the common musket. See Eli Whitney Blake to the Editors, *Daily National Intelligencer* (Washington), July 8, 1835; Asa H. Waters, “Thomas Blanchard, The Inventor,” *Harper’s New Monthly Magazine* 63 (1881): 259–60.

³⁶George H. Daniels, “The Big Questions in the History of American Technology,” *Technology and Culture* 11 (January 1970): 13–14; Paul J. Uselding, “Henry Burden and the Question of Anglo-American Technological Transfer in the Nineteenth Century,” *Journal of Economic History* 30 (1970): 317.

³⁷John H. Hall to Lewis Cass, August 4, 1834, RG 156, OCO, Letters Received; Hall to Cass, October 15, 1834, *ibid.*; Hall to Statira P. Hall, October 15, 1834, Hall-Marmion Letters, Harpers Ferry National Historical Park, Harpers Ferry, W.Va.; Hall to Cass, March 18, 1835, RG 156, OCO, Special File, Correspondence Relating to Inventions; William A. Hall to House Committee on Military Affairs, February 7, 1840, RG 233, HR, Petitions, HR26A.



FIG. 4.—Model 1819 Hall rifle manufactured by Simeon North in 1832. Four years earlier North had received a government contract for 5,000 Hall rifles. Using precision techniques, he succeeded by 1832 in making weapons interchangeable with those made by Hall at Harpers Ferry. The achievement of interarmory standards for uniform production marked a significant technological advance which laid the basis for what eventually became known abroad as the “American System” of manufacture. (Courtesy of the Fuller Gun Museum, Chickamauga and Chattanooga National Military Park, Fort Oglethorpe, Georgia.)

allegations, and, equally significant, the chief of ordnance and chairmen of the House and Senate committees on military affairs confirmed their validity.³⁸

Surely ample opportunity existed for assessing and copying Hall's methods. Between 1828 and 1835 numerous gun makers visited Harpers Ferry. Among the most prominent were Simeon North, Lemuel Pomeroy, Asa Waters, and Marine T. Wickham. Springfield's Roswell Lee, having frequently inspected the rifle works and served as acting superintendent at Harpers Ferry on two separate occasions, also became intimately acquainted with the inventor and his machinery. In 1826 he not only hosted the Carrington committee but also helped select its members.³⁹ Since Hall's contracts implicitly guaranteed the free use of his inventions at both national armories, Lee had ready access to his associate's drawings and patterns.⁴⁰ Finally, as the

³⁸Colonel George Bomford to Lewis Cass, September 16, 1834, RG 156, OCO, Letters Sent to the Secretary of War; Bomford to James I. McKay, February 8, 1836, RG 156, OCO, Letters Sent; Colonel George Talcott to Joel R. Poinsett, March 17, 1840, RG 156, OCO, Letters Sent to the Secretary of War; J. W. Allen, Report on the Petition of John H. Hall, April 25, 1840, RG 233, HR, Petitions, HR26A; W. S. Archer to Secretary of War, May 10, 1842, RG 156, OCO, Letters Received; Isaac N. Coffin to E. Stanley, June 6, 1842, RG 233, HR, Petitions, HR27A.

³⁹Major William Wade to Roswell Lee, October 9, 1826, RG 156, SAR, Letters Received; Lee to Wade, October 17, 1826, RG 156, SAR, Letters Sent; Lee to Wade, October 20, 1826, *ibid.*; Lee to James Carrington, October 17, 1826, *ibid.*

⁴⁰Colonel George Talcott to Joel R. Poinsett, March 17, 1840, OCO.

Ordnance Department encouraged the use of Hall's machinery by others and possessed a complete set of specifications, it also served as a transmission center, a point for the diffusion of technical information.⁴¹

There were, in addition, other equally effective means by which Hall's improvements reached the Connecticut Valley. One likely agent is James Carrington. As chairman of the congressional committee investigating the rifle works at Harpers Ferry he possessed firsthand knowledge of the subject, and, judging from the enthusiastic language of his report, he exhibited more than just passing interest in Hall's cutting machinery.⁴² Moreover, his role as a transmitter is all the more credible in light of recent research on Eli Whitney and the milling machine.

In challenging the great-man theory of history, Edwin A. Battison not only refutes the legendary belief that Eli Whitney invented the modern mode of milling iron but also demonstrates conclusively that a machine used at Whitney's "Mill Rock" factory "must have been made after Whitney's death in 1825."⁴³ Battison's further research indicates that the so-called single "Whitney" machine reveals several important features possessed by Hall's cutting engines. Of particular interest is the use of a self-acting carriage which disengages its screw feed by means of a spring latch once the work traverses the cutter.⁴⁴ Naturally it would be presumptuous to assume that the existent "Whitney" machine represents a duplicate copy of Hall's straight cutting engine. Nevertheless, Battison sets forth the distinct possibility that this machine was built from sketches and information supplied by James Carrington to his former employers upon returning to Connecticut in January 1827.⁴⁵

Several of Hall's mechanics also carried technical information to the Connecticut Valley. In June 1827 two of his three principal workmen departed Harpers Ferry for New England. Soon afterward, Hall

⁴¹John H. Hall to Colonel George Bomford, February 9, 1826, RG 156, OCO, Special File, Correspondence Relating to Inventions; Hall to Statira P. Hall, March 8, 1835, Hall-Marmion Letters; List of Drawings, Plates, Maps and Illustrations of Ordnance Installations and Equipment in the Office of the Chief of Ordnance, 1817-1890, RG 156, OCO.

⁴²Carrington Committee Report, January 6, 1827, OCO.

⁴³Battison (n. 3 above), p. 33. For a perceptive treatment of Whitney's early career see Robert S. Woodbury, "The Legend of Eli Whitney and Interchangeable Parts," *Technology and Culture* 1 (1960): 235-53.

⁴⁴Edwin A. Battison, "A New Look at the 'Whitney' Milling Machine," *Technology and Culture* 14 (October 1973): 592-98.

⁴⁵*Ibid.*, p. 596. Although Carrington resigned as Whitney's foreman in 1825, he continued to serve in the capacity as an inspector of contract muskets at Whitneyville until his retirement in 1830.

reported, "Three others upon whom I relied most for the correct execution of the arms, left me also, as I would not comply with their unreasonable demands for an increase of wages, which they would never have made but for the departure of the others."⁴⁶ Armory payrolls reveal that William Robinson, Jacob B. Mong, Samuel Brantree, Lenox Compton, and Elizur B. Cogsill terminated employment at the rifle works during the summer of 1827.⁴⁷ Of this number Robinson, Mong, and Brantree served as full-time machinists, Compton as a stocker, and Cogsill as a screw maker. Although their exact destinations are unknown, it appears that one or more of them accompanied Roswell Lee to Springfield when the latter ended his duties as acting superintendent at Harpers Ferry in June 1827.⁴⁸ In any event, Lee's correspondence indicates that he employed a number of former Hall workmen. In 1830, for instance, he acquired the services of an able machinist named John Norman.⁴⁹ By 1832 he had added two more armorers to the roster at Springfield.⁵⁰ However, by far the most notable person to leave Hall and go to New England was Nathaniel French.

French's career exemplifies the roving tendencies of early American mechanics. Of unknown origin and background, his name first appeared on Hall's payroll in January 1821. The man undoubtedly possessed prior experience and ability because he received the relatively high wage of \$1.75 per day, a sum second only to the earning power of Hall's shop foreman, Robert Blanchard, and a forger named Timothy Herrington.⁵¹ Employed as a pattern maker and machinist, French remained at the rifle works up through the summer of 1827. But evidently the promise of higher wages lured him away, for he accepted the position as master machinist under Simeon North and assisted the latter in introducing the manufacture of Hall rifles at

⁴⁶John H. Hall to Colonel George Bomford, June 25, 1827, RG 156, OCO, Letters Received; Hall to Bomford, December 10, 1827, *ibid.*

⁴⁷Payrolls and Accounts of William Broadus, Harpers Ferry, July 16, 1827, Record Group 217, Records of the U.S. General Accounting Office, Second Auditor's Accounts, Acct. no. 10876 (hereafter cited as RG 217, GAO); Payrolls and Accounts of William Broadus, October 1, 1827, *ibid.*, Acct. no. 11127; Payrolls and Accounts of William Broadus, March 7, 1828, *ibid.*, Acct. no. 6194.

⁴⁸James Stubblefield to Roswell Lee, June 15, 1827, RG 156, SAR, Letters Received.

⁴⁹John H. Hall to Colonel George Bomford, September 20, 1830, RG 156, OCO, Letters Received; Hall to Bomford, October 12, 1830, *ibid.*; Bomford to Roswell Lee, October 29, 1830, RG 156, OCO, Letters Sent; Bomford to John Norman, October 29, 1830, *ibid.*

⁵⁰Robert Blanchard and Nahum W. Patch. Both men resided at Springfield while serving as contract arms inspectors at Middletown.

⁵¹Payrolls and Accounts of William Little, Harpers Ferry, May 10, 1821, RG 217, GAO, Second Auditor's Accounts, Acct. no. 4830.

Middletown.⁵² After spending two years at North's factory, French again packed his bags and returned to Harpers Ferry for another sojourn at Hall's works. Finally, at the summons of Roswell Lee, he took a permanent position at the Springfield armory in May 1831.⁵³ The implication is clear: in less than five years Nathaniel French plied his trade at three leading arms-making establishments. The mechanical information he carried from one armory to another cannot be dismissed lightly, for here was fresh know-how and the ability to execute it embodied in one person.

* * *

In the meantime, a third stage in the evolution of the milling machine occurred at North's plant in Middletown. Essentially it consisted of mounting the cutter spindle in upright sliding journal boxes so that, by means of screw adjustment, its position could be raised or lowered in relation to the type of work being done. This innovation substantially increased the machine's versatility. Yet, interestingly enough, the development took place after Nathaniel French left Middletown in 1829. Writing to Charles H. Fitch in 1882, William Smith recounted that, upon French's departure, North's superintendent appointed "Mr. William Ferre and myself jointly as master machinists, and advanced the pay of each of us, and we got up the milling machine with the sliding spindle boxes." Furthermore, Smith noted, "we determined to get up a machine that would meet all requirements as far as possible, and it has stood forty-six years without material improvement."⁵⁴ Assuming that Smith's statement is correct and that Hall's improvements had been introduced at Middletown, it appears that all the essential elements of the modern milling machine existed at Simeon North's factory no later than 1831 or 1832.

Smith's testimony bears up remarkably well under scrutiny. It is known, for instance, that the Springfield armory had several machines in operation for milling lock components and mounting for muskets by 1832.⁵⁵ Apparently built by Nathaniel French and

⁵²Payrolls and Accounts of William Broadus, Harpers Ferry, July 16, 1827, Acct. no. 10876, GAO; Fitch (n. 15 above), p. 526.

⁵³Payrolls and Accounts of William Broadus, Harpers Ferry, December 16, 1829, RG 217, GAO, Second Auditor's Accounts, Acct. no. 13626; Nathaniel French to Roswell Lee, March 7, 1831, RG 156, SAR, Letters Received; Lee to French, March 14, 1831, RG 156, SAR, Letters Sent; French to Lee, March 22, 1831, RG 156, SAR, Letters Received; Lee to French, April 5, 1831, RG 156, SAR, Letters Sent.

⁵⁴Fitch, "The Rise of a Mechanical Ideal" (n. 15 above), p. 526; Fitch, *Tenth Census of the United States: Manufactures* (n. 21 above), 2:641.

⁵⁵George Rust, Jr., to Colonel George Bomford, March 3, 1832, RG 156, OCO, Letters Received.

Thomas Warner, there is nothing to suggest that they possessed any means for vertical adjustment. Similarly, in his report on interchangeable manufacture, Charles H. Fitch identifies a machine used at Springfield in 1834 for flat-milling monkey wrenches as a fixed-spindle type.⁵⁶

Certainly officials at the Springfield armory acknowledged Middletown as a leading machine-building center.⁵⁷ On numerous occasions during the 1830s Roswell Lee and his successor, John Robb, dispatched workmen downriver to observe various mechanical improvements. In December 1831, for example, Lee instructed Nathaniel French to examine and make drawings of certain machines used by Robert Johnson, Simeon North, and Wilcox & Markham for the purpose of introducing them at Springfield.⁵⁸ Likewise, Robb sent his master armorer, Elizur Bates, and a mechanic named Calvin Barker to Middletown in April 1834 to "procure castings for a milling Engine."⁵⁹ According to Fitch, this machine had a vertical spindle adjustment and supplied the model for a plain miller built around 1835 by Thomas Warner of Springfield for making lock plates of uniform thickness. Soon afterward the Springfield armory constructed another machine for milling the irregular edges of lock plates on a design previously introduced by Robbins and Flagg at the Waters armory in Millbury, Massachusetts.⁶⁰ The decisive line of influence, however, emanated from Middletown. Indeed, the national armory not only purchased castings in that community but recruited its workmen there as well. Among the most prominent was Simeon North's master machinist, William Smith, who moved to Springfield in 1837.⁶¹

With Smith, Nathaniel French, Cyrus Buckland, and Thomas Warner employed under the same roof, few if any manufacturing

⁵⁶Fitch, *Tenth Census of the United States: Manufactures*, 2:641.

⁵⁷See, for example, Lieutenant Daniel Tyler to Colonel George Bomford, November 30, 1830, RG 156, OCO, Letters Received.

⁵⁸Roswell Lee to Simeon North, Robert Johnson, and Markham & Wilcox, December 24, 1831, RG 156, SAR, Letters Sent; Roswell Lee to Nathaniel French, December 24, 1831, *ibid.*; Mrs. Zoa French to Lee, January 24, 1832, RG 156, SAR, Letters Received.

⁵⁹John Robb to Elizur Bates, April 17, 1834, RG 156, SAR, Letters Sent; Samuel Sizer to Robb, June 28, 1834, RG 156, SAR, Letters Received. Rather than suggesting the transfer of mechanical ideas solely by itinerant mechanics, these episodes indicate that the armory masters themselves not only kept abreast of novel techniques successfully introduced elsewhere, but actively sought them out by directing borrowing from above. I am indebted to Paul Uselding for pointing this out in an earlier draft of this paper.

⁶⁰Fitch, *Tenth Census of the United States: Manufactures*, 2:641.

⁶¹Fitch, "The Rise of a Mechanical Ideal" (n. 15 above), p. 525.

establishments in the United States could have equaled Springfield for sheer mechanical talent between 1837 and 1842. Their very presence goes far to explain the accelerated rate of machine building that occurred at the armory during these years, culminating in 1844 with the production of the first conventional musket made entirely from interchangeable parts. Yet, even though Cyrus Buckland gained lasting fame for his more streamlined versions of Thomas Blanchard's stocking machinery and other mechanical innovations, it is often overlooked that Warner, assisted by French and Smith, made outstanding contributions to plain milling. By combining the use of formed cutters, vertical spindle adjustment, power feed, and self-acting disengagement gearing, these "Springfield" machines represented a cumulative synthesis of principles previously introduced by Simeon North, John H. Hall, and countless others.⁶² Equally important but curiously forgotten, they supplied the prototype for the first commercially produced plain milling machines devised during the 1840s by Jacob Corey MacFarland of the Ames Manufacturing Company and Frederick W. Howe of Robbins & Lawrence.⁶³ The

⁶²From an interpretative point of view, the process described in this paper coincides with Abbott P. Usher's analysis of the emergence of novelty as set forth in his classic study, *A History of Mechanical Inventions*, rev. ed. (Cambridge, Mass., 1966), pp. 56–83.

⁶³For a discussion of these machines, see Edward G. Parkhurst, "More Early Milling Machines," *American Machinist*, June 28, 1900, pp. 605–6.

The Ames Manufacturing Company of Chicopee, Massachusetts, deserves an important place in the annals of American industry. Organized in 1834 by Nathan P. and James T. Ames, it became one of the earliest firms in the United States to manufacture and sell a standard line of machine tools to the general public. Although the Ames brothers marketed everything from Boyden water turbines to gun stocking machinery, one of the most popular items in their repertory were plain milling machines. During the 1840s and 1850s workmen commonly referred to these machines as "M'Farland" millers. The term refers to Jacob Corey MacFarland, a mechanic who received his early training at the Springfield armory and served as foreman of Ames' machine shop between 1845 and 1858.

In view of his background, MacFarland probably worked with Thomas Warner, Nathaniel French, and William Smith in developing the milling machine at Springfield during the late 1830s. More than likely he added a few embellishments of his own. Whatever the case, it is clear that the earliest plain millers made at the Ames works were based on patterns owned by the Springfield armory. Furthermore, as Robbins, Kendall & Lawrence of Windsor, Vermont, initially equipped their plant with Ames-built machinery in 1845 and made abundant use of Springfield patterns thereafter, it appears that the same observation holds true for Frederick W. Howe's famous plain milling machine of 1848. For information on these subjects, see Felicia J. Deyrup, *Arms Makers of the Connecticut Valley* (Northampton, Mass., 1948), pp. 125–26; Vera Shlakman, *Economic History of a Factory Town* (Northampton, Mass., 1935), pp. 27, 81–86; "History of a Famous Industry: Ames Company's Long Career," *Springfield (Mass.) Republican*, January 15, 1899, p. 10; Martin Lovering, *History of the Town of Holland, Massachusetts* (Rutland, Vt., 1915), pp. 604–7; Nathan P. Ames to Benja-

latter, in turn, provided inspiration for the famous Lincoln-type machines made at Hartford, Connecticut, during the 1850s. Beyond this point, the general design of plain milling machines changed very little up through the end of the 19th century.⁶⁴

* * *

Though much research remains to be done on the origins of the American System, several conclusions appear warranted at this juncture. Those who view technological change in Schumpeterian terms might profess to see the contributions of Simeon North or John H. Hall as dramatic examples of personal ingenuity and as additional

min Moore, January 24, 1845, N. P. Ames Letterbook, James T. Ames Papers, private collection, Cromwell, Connecticut; Colonel George Talcott to Robbins, Kendall & Lawrence, April 10, 1845, RG 156, OCO, Letters Sent; Robbins, Kendall & Lawrence to Talcott, March 10, 1845, RG 156, OCO, Letters Received; Major James W. Ripley to Robbins, Kendall & Lawrence, July 21, 1847, RG 156, SAR, Letters Sent; James S. Whitney to Ames Manufacturing Company, November 28, 1855, *ibid.*

⁶⁴Joseph W. Roe, *English and American Tool Builders* (New Haven, Conn., 1916), pp. 165–66; W. Steeds, *A History of Machine Tools, 1700–1910* (London, 1969), pp. 73–74, 102; Robert S. Woodbury, *History of the Milling Machine* (Cambridge, Mass., 1960), pp. 33–42.

Numerous uncertainties pervade the subject when one turns to the types of cutters employed on early milling machines. For one thing, the absence of adequate artifact and archival material severely constrains historical knowledge of tool design. It is nonetheless clear that the earliest machines were fitted with cutting tools shaped like rotary files. With the exception of Simeon North's first machine, they also executed contoured work. The record, however, is inconclusive as to when toothed cutters of the modern variety first appeared in machine shops. More than likely they followed an evolutionary pattern of development similar to that of the milling machine as workmen familiarized themselves with the tolerances of tool steel and mastered other technical problems. Be that as it may, there is evidence to suggest the use of coarser toothed cutters at an early date. Several references indicate, for instance, that "saws" capable of removing a substantial chip of iron were employed on John H. Hall's cutting machines. Cutters of similar dimensions were probably used on the "Whitney" machine and Nathan Starr's milling engine of 1829. It is noteworthy that Starr's engine had three stationary "poppet heads," one of which evidently supported the outer end of the cutter spindle. Though it cannot be corroborated directly, the addition of such spindle support suggests the use of mill cutters capable of doing heavier, more demanding work than would normally be required of rotary files. Initially considered, then, toothed cutters may well have appeared in the firearms industry much earlier than is generally recognized. With the advent of standard "manufacturing" milling machines around mid-century, cutting tools of the modern type came into widespread use. However, as their upkeep involved much time and expense, the art of tool making entered a new phase of development in 1864 when Joseph R. Brown of Providence, Rhode Island patented a formed cutter that could be expeditiously ground and sharpened without altering its original shape. Edwin A. Battison provided valuable assistance in interpreting the technical significance of the outboard spindle support on Starr's milling machine.

evidence for the heroic theory of invention. Yet, most mechanical improvements introduced at public and private armories during the early 19th century accrued through a process of modifying previously existing principles and techniques. The evolution of the milling machine clearly illustrates this tendency. It did not spring full blown from the mind of any one person. Rather it took form gradually through a remarkable process of cooperation, transfer, and convergence. Beginning around 1816 a continuous series of innovations in milling spread from Middletown to Harpers Ferry; from Harpers Ferry to Middletown, New Haven, and Springfield; from Middletown to Springfield; and, finally, from Springfield to Chicopee, Windsor, and Hartford. Simeon North and John H. Hall definitely played key roles in this broadly based development. So too did Roswell Lee. But their names must always be mentioned in conjunction with other imaginative mechanics who left little or no written record of their achievements yet contributed substantially to the emergence of modern milling practice.

Some persons, of course, figure more conspicuously as innovators than others. Indeed there is good reason to heed the suggestions of scholars who stress the importance of studying technology as the history of expanding systems.⁶⁵ In this context due emphasis must be placed on the role of mechanical entrepreneurs, for the really significant achievements in technology are not isolated instances of invention but the synthesis of disparate techniques into efficient systems of production. Certainly the American System would never have come into being without the judicious integration of men, machines, and precision measurement methods. It is in this capacity that men such as Simeon North, John H. Hall, and Roswell Lee made their greatest contributions. Nevertheless, it is equally clear that they did not work alone. In a society lacking adequate educational facilities, professional associations, and technical publications, a great deal depended upon the personal transmission and dissemination of mechanical ideas. Time and again factory masters received valuable assistance from itinerant mechanics. Time and again those same mechanics carried away know-how and placed it at the disposal of others. Clearly the study of geographical groupings of mechanics and the movements of individuals within those groups suggest both the wondrous complexity and dynamism of technological change in early industrial America.

⁶⁵See, for example, Hugh G. J. Aitken, ed., *Explorations in Enterprise* (Cambridge, Mass., 1965); and Nathan Rosenberg, "Technological Change in the Machine Tool Industry, 1840–1910," *Journal of Economic History* 23 (1963): 414–43.